Oxygen Therapy and Humidification

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NHS Guy's and St Thomas' NHS Foundation Trust

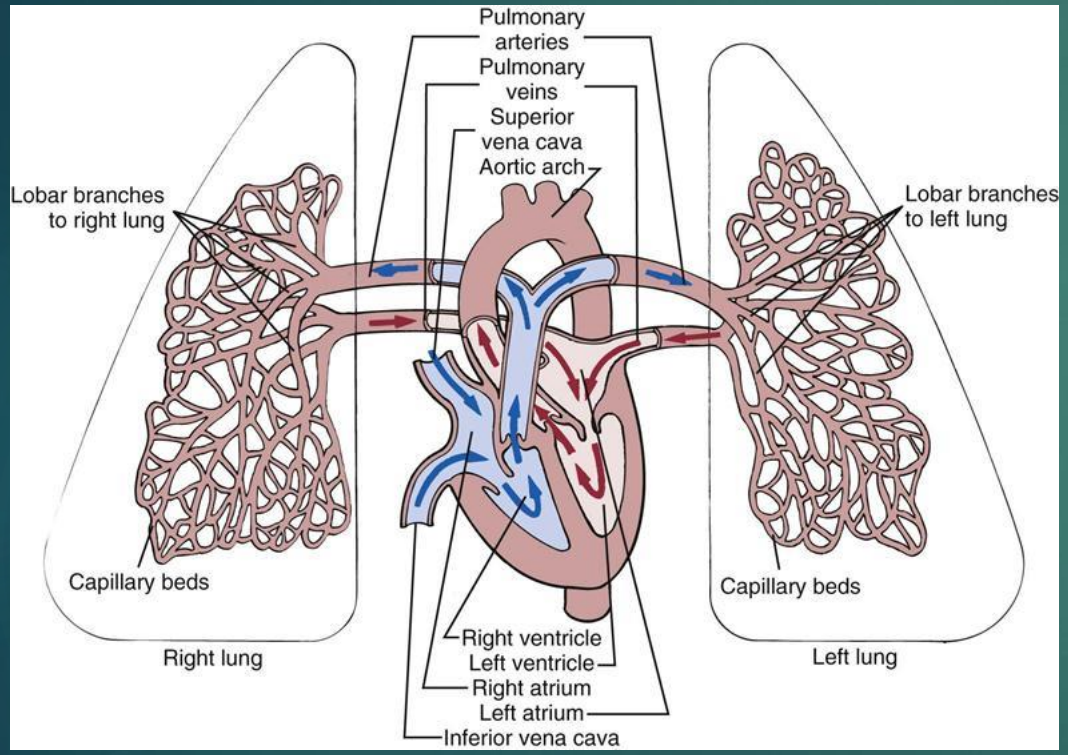
NHS Royal Brompton and Harefield hospitals

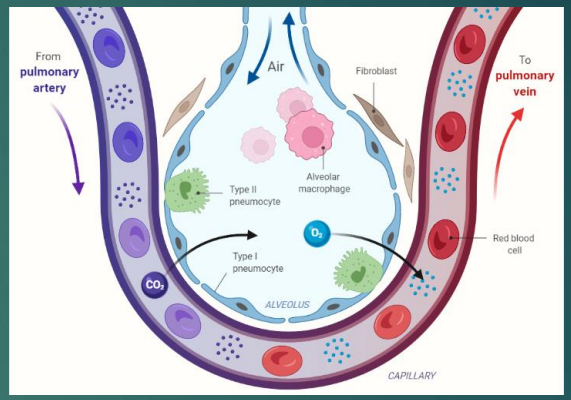
Learning Objectives

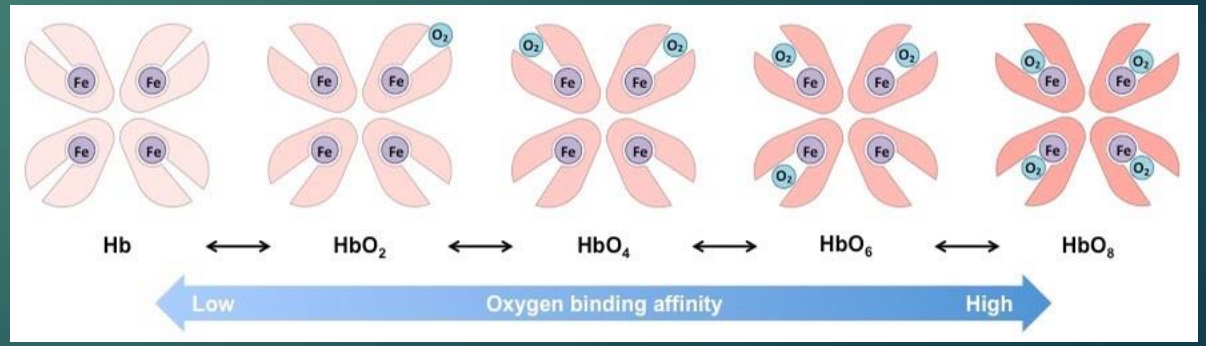
- Physiology of oxygenation
- Normal values of oxygenation and ABG
- Type 1 and type 2 respiratory failure
- Medical oxygen delivery systems
- Humidification of the airway
- Target saturations



Oxygenation and circulation





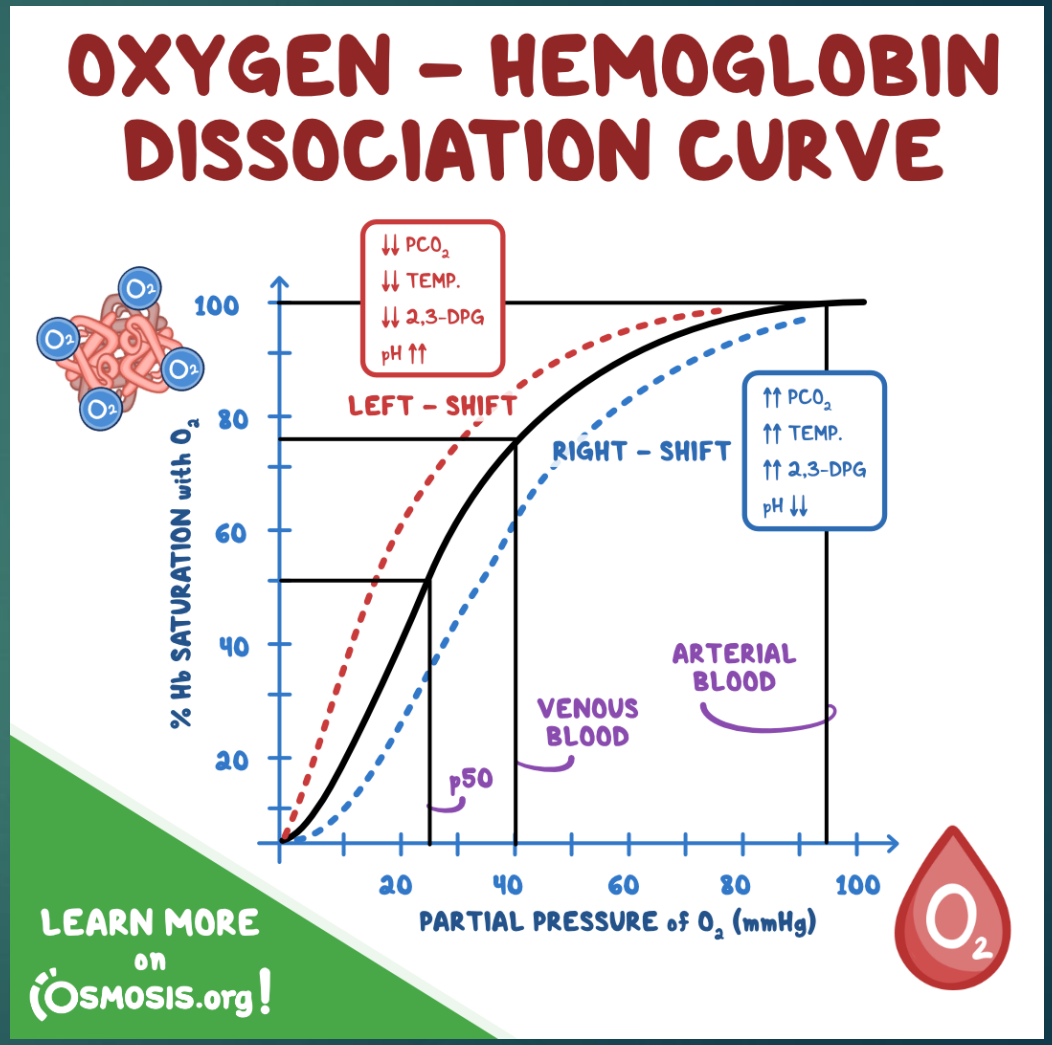


Let's get the terminologies right

- SO2
  - The percentage of haemoglobin in the blood that is bound with oxygen
  - Representing oxygen-carrying capacity and oxygen delivery efficiency
  - Can only be measured via blood sample
  - Much more accurate but difficult to access
- SpO2
  - Same as SO2 but measured peripherally and non-invasively via a pulse oximetry
  - Can be slightly less accurate due to factors such as blood flow, nail polish and movements
- PO2
  - Partial pressure of dissolved oxygen free flowing in blood, not binded to haemoglobin
  - Represents oxygenation of blood from lungs, and amount of oxygen available to be utilized
- FiO2
  - Fraction of inspired oxygen
  - Percentage of oxygen in inhaled gas
  - Accurate way of measure exact amount of oxygen delivered into body
- Oxygen flow
  - Usually use to describe the speed of 100% oxygen delivered through a flow metre from source
  - Measured in L/min
- Hypoxia
  - Deficiency of oxygen in tissues
- Hypoxaemia
  - Deficiency of oxygen in blood

Normal values

- SpO2 / SO2
  - 94% or above in an uncompromised individual
- PaO2
  - 10-13.3kpa
- Oxy-haemoglobin dissociation curve
  - Cooperative binding
  - High PO2 supports high affinity of O2 binding
  - Low PO2 supports faster O2 offloading
  - Bohr and Haldane effect



Normal ABG Values

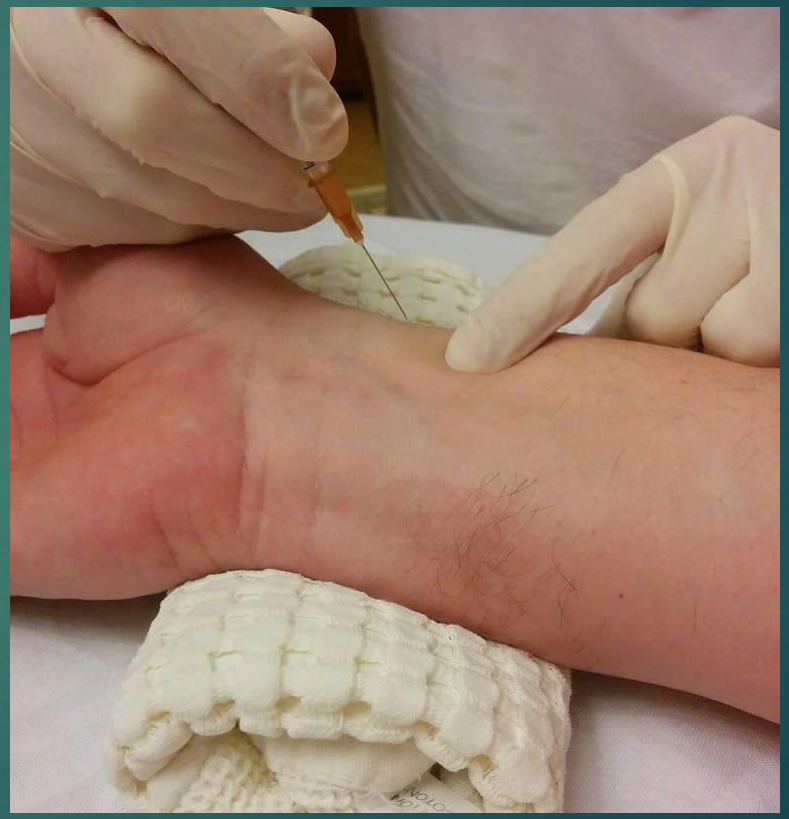
PH : 7.35 - 7.45

PaCO2 : 4.6 - 6 kpa

PaO2: 10 - 13.3 kpa

HCO3 : 22 – 26 mmol/L

BE: -2 - +2 mmol/L



Respiratory failure

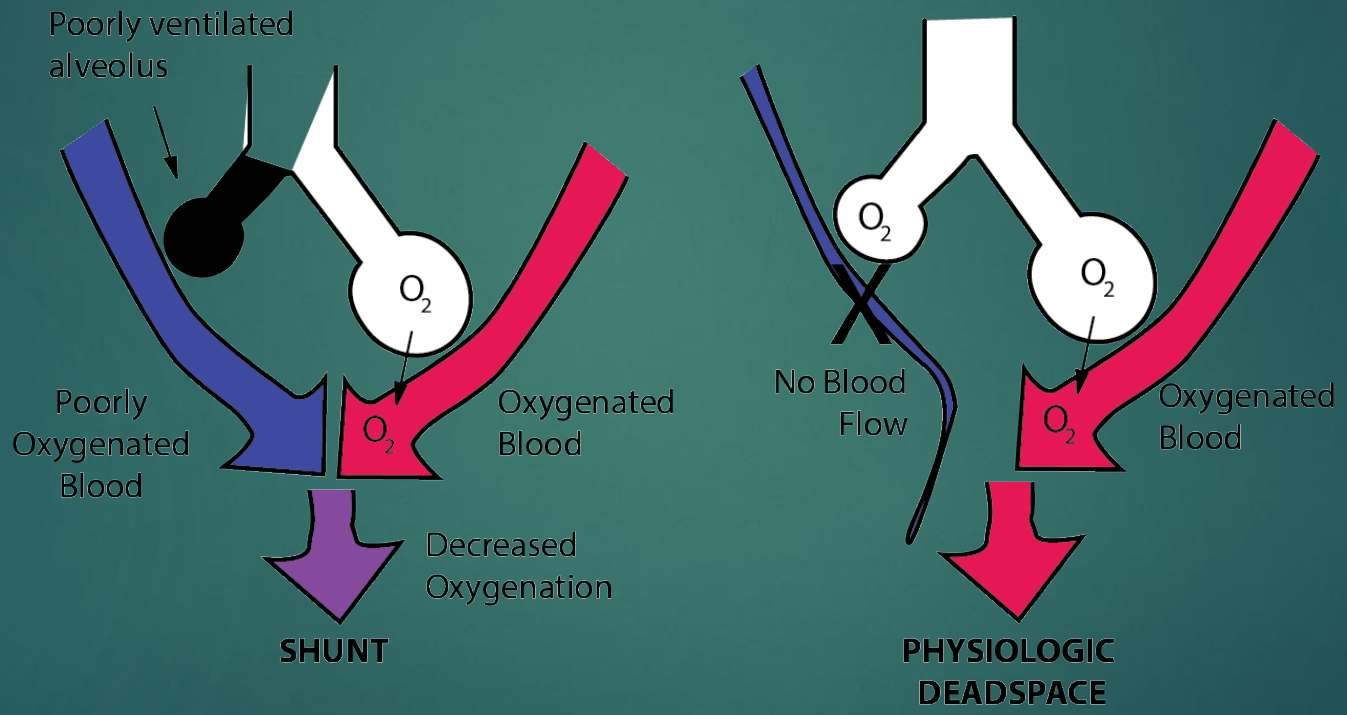
Type 1

- **Low PaO_2**
- Normal $PaCO_2$
- "Hypoxia"
  - Pneumonia
  - Fluid Overload
- when the respiratory system cannot adequately provide oxygen to the body

Type 2

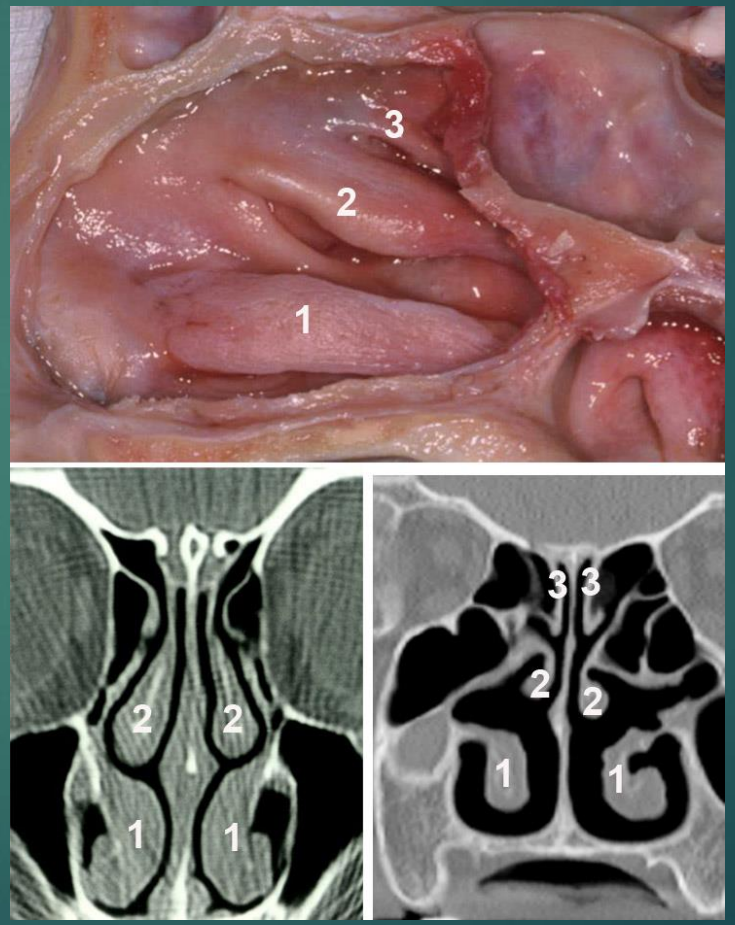
- Low or normal PaO_2
- **High $PaCO_2$**
- "Hypercapnia" +/- Hypoxia
  - Neuromuscular compromise
  - Decreased drive to breath
  - Chest wall disorders
  - Chronic COPD patients
- Poor ventilation of the alveoli leading to retention of Co_2

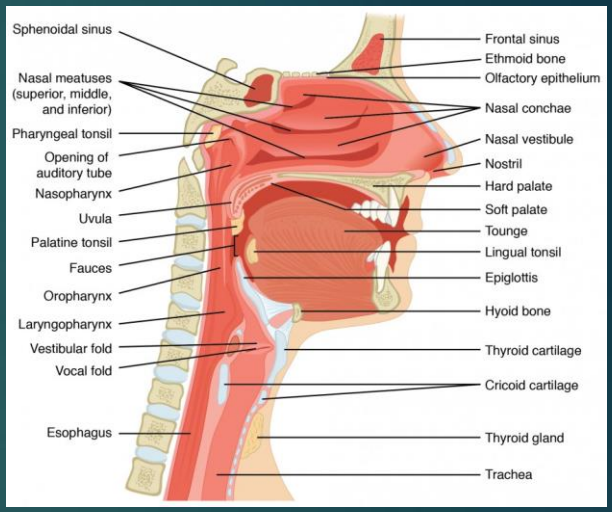
Physiological shunting and deadspace



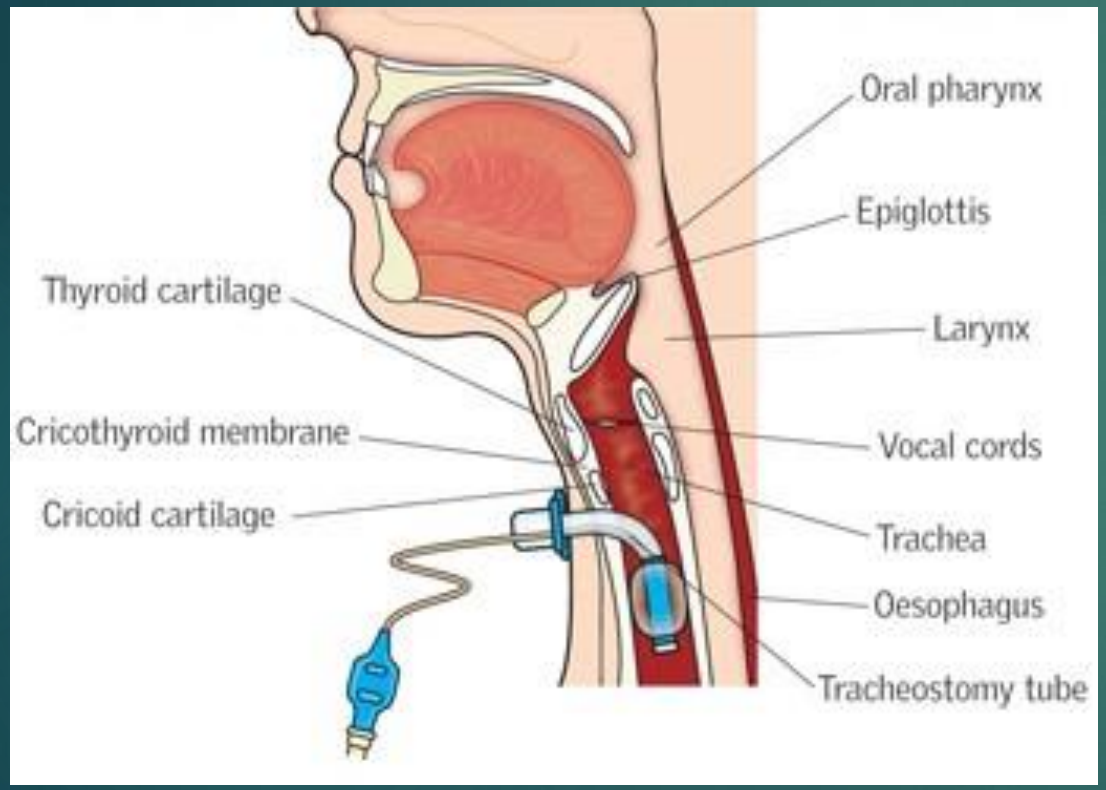
Humidification

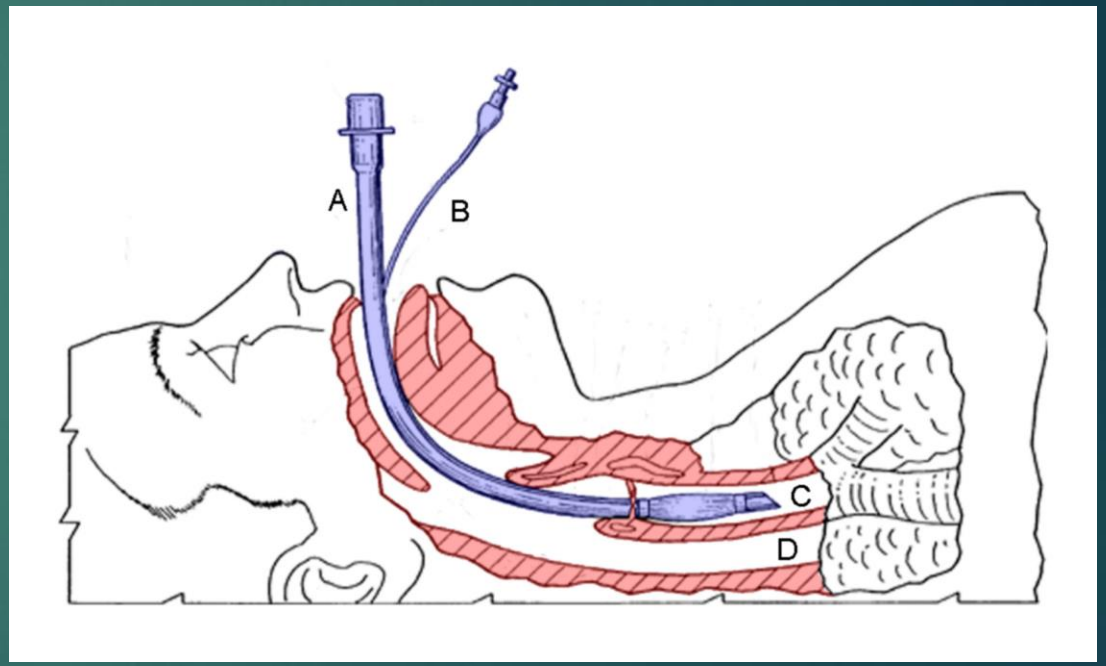
- How :
  - Air flows through nasal turbinates
  - Air is humidified and heated almost 80% via the nose and sinus
- Why :
  - To condition the air in preparation of gas exchange, which happens most ideally at 37 degrees and 100% relative humidity
  - Maintains the mucociliary transport system
    - Prevents drying of secretions
    - Aids secretion clearance





What happens in artificial airways





Now you understand what the body struggled with…How do we manage that?



Page 10 Mason County News

Outdoors

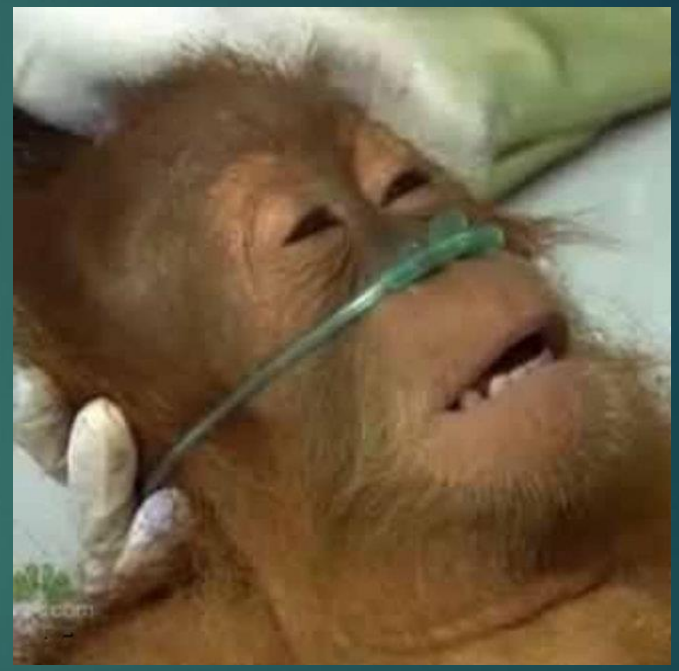
Outdoors Outpost

By KENDAL FLEMMING

Breathing oxygen linked to staying alive

Oxygen support

- Oxygen is a DRUG and must be prescribed
  - Should include target saturation
- Oxygen is a treatment for hypoxaemia, NOT breathlessness
- Oxygen delivery is measured in 2 ways :
  - Litres per minute (LPM)
  - Fraction of inspired oxygen (Fio2)
- Split into 4 categories :
  - Fixed devices
  - Variable devices
  - Low flow
  - High flow devices

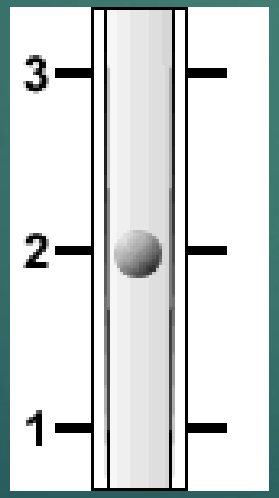


Reading a flow meter

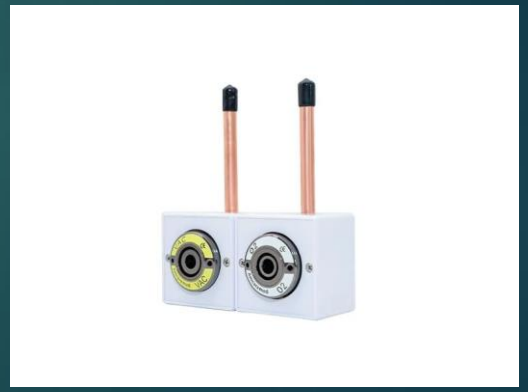
- Read the value that runs through the middle of the black ball
- Make sure it is connected to oxygen port NOT medical air
- Use low flow metre for <1 L/min
- Wall outlet always delivers pure FiO2 100% oxygen











Fixed Vs variable devices

- Variable devices
  - Provide a flow of oxygen but no guaranteed FiO2
  - Measured in L/min
  - FiO2 received depends on
    - Change in RR
    - Nose/mouth breathing
    - Seal/leak
- Fixed devices
  - Provide definite FiO2 based on device chosen
  - Does not vary with RR or even flow (after reaching minimum threshold)

Low flow : Variable devices



- Nasal cannula
  - Delivers 1 - 6 L/min
  - NOT to be used above 6L/min as it can dry up nasal mucosa
  - Pros:
    - Used in hospital and in the community
    - Comfortable
    - Easy to mobilise
    - Eating, drinking and talking are easy
  - Cons
    - Dry oxygen
    - Variable FiO2 –4% per L/min increase roughly



Low Flow : Variable devices

- Simple face mask
  - Can deliver 5-10 L/min
  - Pros:
    - Good for mouth breathers
    - Good if nose not appropriate
  - Cons
    - Affects communication, eating and drinking and mouth care
    - Dry oxygen
    - Inaccurate FiO2 delivery
    - Lower than 5L/min will lead to CO2 re-breathing





Low flow devices : Variable devices

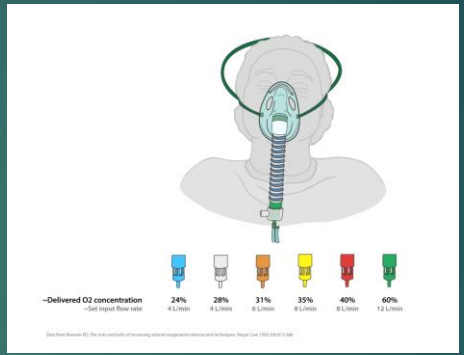
- Non Re-breath mask :
  - Delivers 60-80% FiO2, needs 15L/min of flow
  - Need bag to be completely inflated before applying, and should remained inflated throughout usage
  - Pros:
    - Delivers high amount of O2
    - Great for emergencies : fast, simple set up
  - Cons
    - Dry oxygen
    - Variable fiO2
    - Affects communication, eating etc





Low flow devices : Fixed devices

- Venturi mask
  - Attach onto a face-mask
  - Provide a fixed FiO2 of oxygen as long as it reaches minimum flow requirement
  - Has recommended ideal flow rate for each FiO2 option, but this can be increased in certain circumstances
  - Pros :
    - Good for mobility as will deliver a fixed FiO2 even with a change in respiratory rate
    - Provide high FiO2 with simple set up
    - Can easily monitor oxygen demand
  - Cons
    - Dry Oxygen
    - Affects communication, eating etc





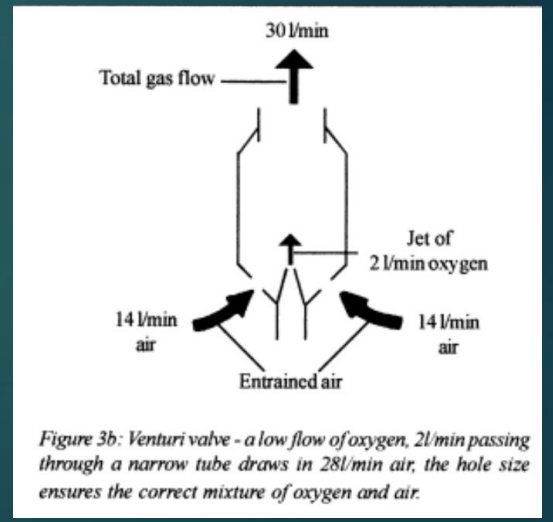


Figure 3b: Venturi valve - a low flow of oxygen, 2l/min passing through a narrow tube draws in 28l/min air, the hole size ensures the correct mixture of oxygen and air.

Total Flow – l/min.

Oxygen Flow l/min.	24	28	31	35	40	60
20					84	40
15				84	62	30
12				67	50	24
10			78	56	41	
8		89	63	45		
6		67	47			
4	102	44				
2	51					

FiO2

Low flow devices : Fixed devices

- Cold humidified / Acqua pack:
  - Fixed Fio2 from : 28%-60%
  - Bubbled through sterile water
  - Pros:
    - Humidification
    - Fixed Fio2
  - Cons:
    - Not portable
    - Very noisy
    - Affects communication, eating etc







High flow devices

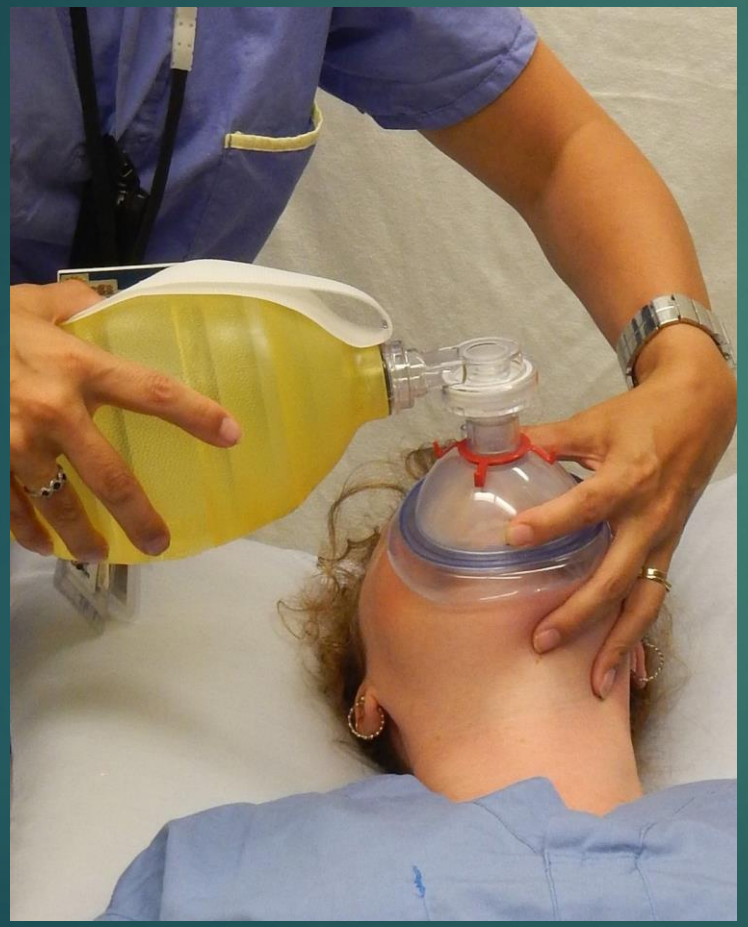
- High flow nasal cannula :
  - Give a fixed Fio2 but also a high flow of Oxygen
  - Can give from 21%-100% FiO2 and up to 60L flow
  - Can be given through Airvo/Optiflow device or ventilator



  - Pros
    - Heated and humidified
    - Relieves work of breathing
    - Provides minimal PEEP – barely beneficial
    - Evidence on supporting central and upper airway secretion clearance only
  - Cons
    - Not good for mouth breathers
    - Can cause pressure areas on face
    - Can be uncomfortable

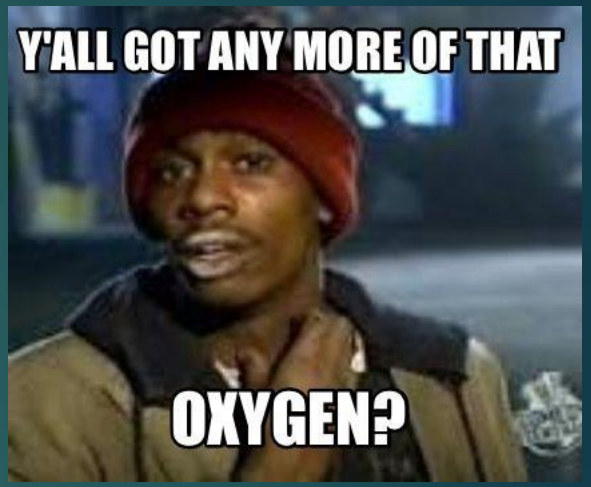


What about these ones?





What if all of the above fails…







Something for another day……..

Now you know what's wrong and what to give… But how much?



Target oxygenations

- Normal:
  - SpO2 94 – 98%
- Asleep:
  - Average SpO2 90%
- CO2 retention:
  - SpO2 88– 92%
  - PO2 >7.3Kpa
- Pulmonary hypertension / Cor pulmonale / Polycythaemia:
  - PO2>8kpa
  - SpO2 >90%
- Paediatric cyanotic congenital heart conditions
  - SpO2 75 - 85%

Oxygen and Hypercapnia

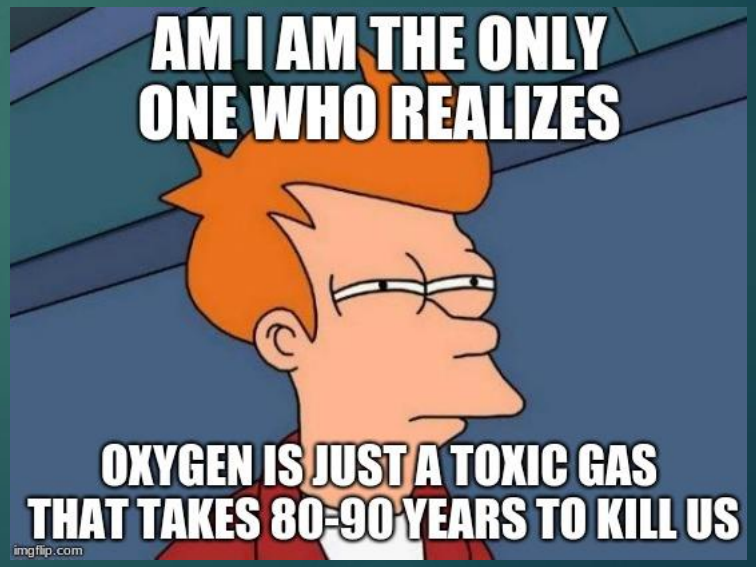
- When we give chronic hypercapnic patients too much oxygen …
  - Some conditions such as COPD leads to chronic CO2 retention
  - Desensitisation of peripheral chemoreceptors
  - Central receptors only able to detect an increase in oxygen levels. This leads to 3 effects and 1 outcome:

  - **Haldene effect** : Hb carrying CO2, dumps into the blood in favour of O2
  - **Reduced drive to breath**
  - **Reversal of hypoxic vasoconstriction :** increased shunt

  - All of these lead to an **increase in PaCO2 which worsen the T2RF**

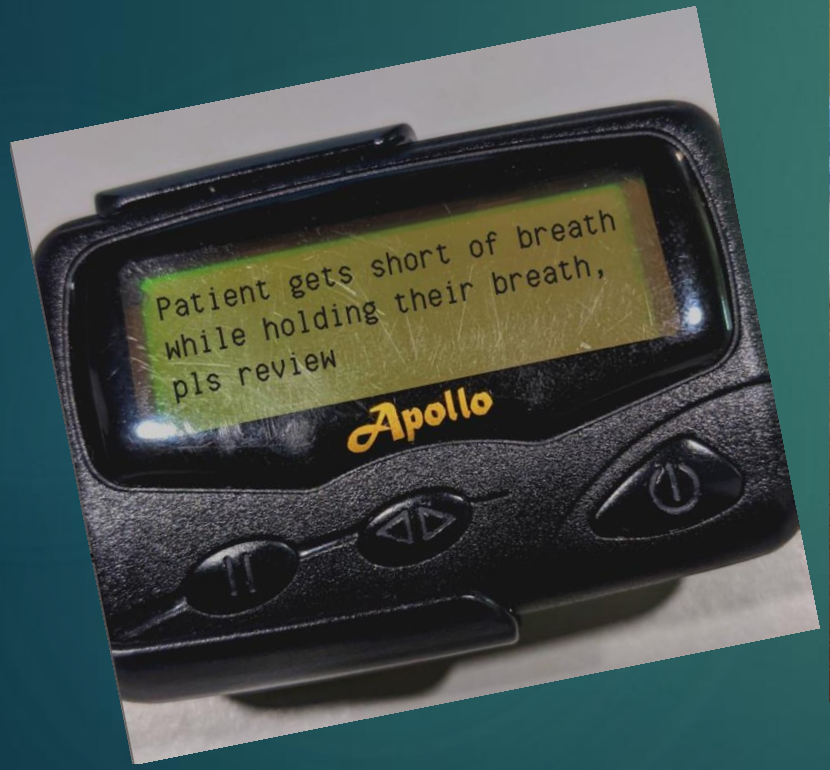
Oxygen toxicity

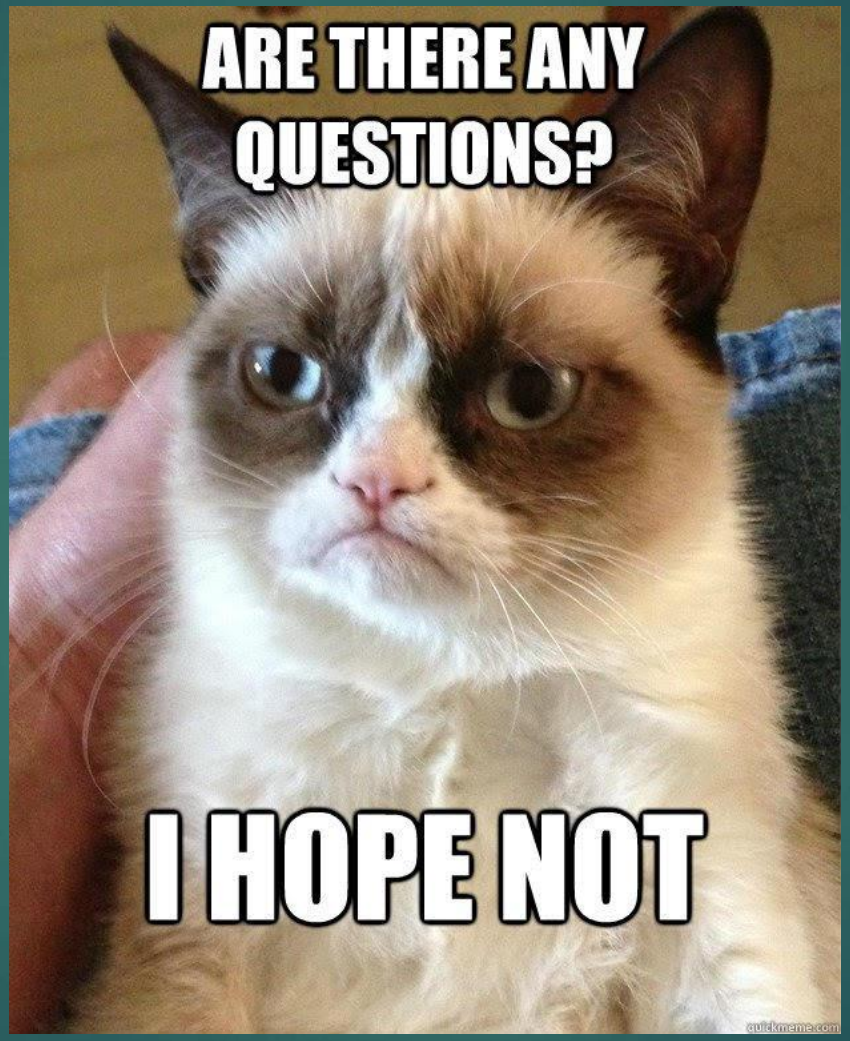
- High level of oxygen leads to production of oxygen derived free radicals
  - Cell damage
  - Acute lung injury secondary to increased permeability
  - Reduced cilia action
  - Impaired macrophage function
  - Reduced surfactant production



Absorption atelectasis

- Created nitrogen washout
  - Nitrogen contributes to majority of the composition of inhaled gas
  - Oxygen is highly soluble to blood in comparison to nitrogen
  - Reduced gas in alveoli to maintain patency


Patient gets short of breath
while holding their breath,
pls review
Apollo


ARE THERE ANY QUESTIONS?
I HOPE NOT


Chief Complaint
Hes not breathing goof